Lecture 1

States, determinism and phase portraits

Objectives. *This course asks one question:* what is a classical mechanical system? *It answers three times, each answer a layer deeper, and the question splits into three: what is a state, what is an observable, and how do they evolve? This lecture gives the first, provisional answers, starting from a question about prediction: what do we need to know about a system now in order to know its future? The answer is the* state*; the states form the* phase space*, and the law of motion becomes a* flow *on it. The lecture states the theorems on differential equations that make this honest, proves that the flow of a particle in a potential bounded below exists for all time, and reads the whole motion of a system with one degree of freedom off the graph of its potential, with the pendulum as the example. It ends with a question the course will answer: what forbids the attractors that friction makes?*

1.1 Determinacy

Newton's principle

A system of N point particles in space is described, at each instant, by the positions $\mathbf{r}_1, \dots, \mathbf{r}_N \in \mathbb{R}^3$. Collect them into one list

$$x = (\mathbf{r}_1, \dots, \mathbf{r}_N) \in \mathbb{R}^n, \qquad n = 3N. \tag{1.1}$$

The point x is the *configuration*, and $\mathbb{R}^n$ the configuration space. Two particles on a line are two moving points in real space but *one* moving point in the plane of configurations: a motion of the whole system is a single curve $x(t)$.

Newton's second law says the accelerations are functions of the positions and velocities (and possibly the time),

$$\ddot{x} = f(x, \dot{x}, t). \tag{1.2}$$

Its physical content is *Newton's principle of determinacy*: positions and velocities at one instant determine the whole motion (Arnold, *Mathematical Methods*, §1). It is an experimental fact, not a theorem.

Definition 1.1 (State, phase space)**.** A *state* of a system is a collection of data at one instant that determines its motion at every other instant at which the motion exists. The set of all states is the *phase space* $\mathcal{M}$. When f in (1.2) does not depend on t (and solutions are unique, § 1.2), a state is the pair

$$z = (x, v) \in \mathcal{M} \subseteq \mathbb{R}^{2n}, \qquad v = \dot{x}. \tag{1.3}$$

Phase space then has *twice* as many dimensions as there are degrees of freedom. If f depends on t, the same (x, v) at different times has different futures (try $\ddot{x} = t$); the clock must join the state, giving the $(2n+1)$-dimensional space of (1.6).

Configuration is not state

Release the oscillator $\ddot{x} = -x$ four times from the same position $x_0 = 1$ with four velocities v_0 (Figure 1.1). The solutions $x(t) = x_0 \cos t + v_0 \sin t$ all pass through $x = -x_0$ at $t = \pi$: in the (t, x) plane the curves cross. In the phase plane (x, v) the same motions run on the circles $x^2 + v^2 = x_0^2 + v_0^2$, which are disjoint. § 1.6 proves this in general.

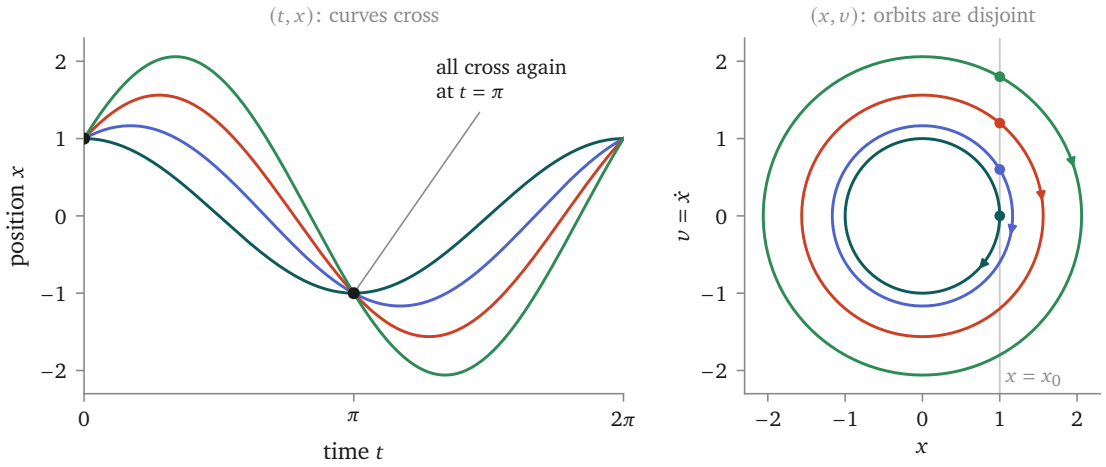


Figure 1.1: The oscillator from one position with four velocities. In (t, x) the curves cross; in the phase plane they are disjoint circles, run clockwise.

Newton as a first-order system

Introducing v as an unknown in its own right turns (1.2) into a system of $2n$ first-order equations,

$$\dot{x} = v, \qquad \dot{v} = f(x, v, t), \qquad \text{that is} \qquad \dot{z} = F(z, t), \quad F(x, v, t) = \big(v,\ f(x, v, t)\big). \tag{1.4}$$

The two systems have the same solutions, $z(t) = (x(t), \dot{x}(t))$. What has changed is the *form*: the law of motion is now a rule assigning to each state the velocity with which it moves — a vector field on phase space.

1.2 Vector fields and flows

Throughout, $U \subseteq \mathbb{R}^m$ is open; for mechanics $m = 2n$ and $U = \mathcal{M}$.

Definition 1.2 (Vector field, integral curve). A *vector field* on U is a map $F : U \to \mathbb{R}^m$ of class C^1, pictured as an arrow $F(z)$ attached at each point z. An *integral curve* (or *solution*) through z_0 is a differentiable curve $\gamma : I \to U$ on an open interval $I \ni 0$ with

$$\dot{\gamma}(t) = F\big(\gamma(t)\big) \quad (t \in I), \qquad \gamma(0) = z_0. \tag{1.5}$$

Such a system is *autonomous*; $\dot z = F(z,t)$ becomes one on $U\times\mathbb{R}$ by promoting time to a coordinate:

$$\dot z = F(z,s), \qquad \dot s = 1. \tag{1.6}$$

Its solutions are those of the original system with $s = t + \text{const}$.

Theorem 1.3 (T1: existence, uniqueness, maximal solutions · *cited*)**.** *Let $F\in C^1(U,\mathbb{R}^m)$. Through each $z_0\in U$ there is a unique* maximal *integral curve, defined on an open interval $(T_-(z_0),T_+(z_0))\ni 0$: every integral curve through z_0 is a restriction of it. Moreover, the* blow-up alternative *holds: if $T_+(z_0)<\infty$, then the solution eventually leaves every compact set $K\subset U$, that is, there is $t_K<T_+$ with $\gamma(t)\notin K$ for all $t\in(t_K,T_+)$. The same holds for T_- as t decreases.*

(Teschl, *Ordinary Differential Equations*, Thms 2.2, 2.13, Cor. 2.16.) The blow-up alternative is the tool we use most: to show that a solution exists for all time, trap it in a compact set.

Example 1.4 (What T1 excludes)**.** $\dot x = 3x^{2/3}$ is continuous but not Lipschitz at 0. Through $x(0)=0$ pass the zero solution and, for each $c\ge 0$, the curve equal to 0 for $t\le c$ and to $(t-c)^3$ after. The present no longer determines the future. This is why every law of motion in the course is at least C^1.

When the maximal solutions are collected together, they form the flow.

Theorem 1.5 (T2: the flow · *cited*)**.** *Let $F\in C^k(U,\mathbb{R}^m)$, $k\ge 1$, and let $\varphi_t(z)$ denote the value at time t of the maximal solution through z. The set $W=\{(t,z) : T_-(z)<t<T_+(z)\}$ is open, the map $(t,z)\mapsto\varphi_t(z)$ is C^k on W, and wherever both sides are defined*

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{t+s} = \varphi_t\circ\varphi_s. \tag{1.7}$$

Each φ_t is a C^k diffeomorphism from its domain onto its image, with inverse φ_{-t}.

(Teschl, Thm 6.1.) The group law is determinacy in formulas: the state at time s determines everything after it. The *orbit* of z is the set $\{\varphi_t(z)\}$ of states it passes through, with the time forgotten.

Corollary 1.6 (Orbits never meet)**.** *Two orbits of an autonomous vector field either coincide or are disjoint.*

Proof. Suppose $\varphi_{t_1}(z_1)=\varphi_{t_2}(z_2)=w$. The curve $t\mapsto\varphi_{t+t_1}(z_1)$ is an integral curve of F through w at $t=0$, because F does not depend on t, so a time-shifted solution is again a solution. So is $t\mapsto\varphi_{t+t_2}(z_2)$. By uniqueness in T1 they are the same maximal curve, so the two orbits are the same set. ∎

Autonomy is used once, and for $\dot z = F(z,t)$ the statement holds only in $U\times\mathbb{R}$. Every orbit is *a point* (an equilibrium, $F(z)=0$), *a closed curve* (a periodic orbit), or *an injective curve* (Teschl, §6.2).

One further import is needed once flows are differentiated, from Lecture 6 on.

Theorem 1.7 (T5: smooth dependence, the variational equation · *cited*)**.** *If $F\in C^k$, the flow is C^k in (t,z), and its Jacobian matrix $M(t)=D\varphi_t(z)$ solves the linear* variational equation

$$\dot M(t) = DF\big(\varphi_t(z)\big)\,M(t), \qquad M(0)=\mathbb{1} \tag{1.8}$$

(Teschl, Thm 2.10).

1.3 Completeness

Definition 1.8 (Complete vector field)**.** F is *complete* if every maximal solution is defined for all $t \in \mathbb{R}$. Then $W = \mathbb{R} \times U$, and $\{\varphi_t\}_{t\in\mathbb{R}}$ is a *one-parameter group* of diffeomorphisms of U.

Completeness makes each φ_t a map $\mathcal{M} \to \mathcal{M}$, and almost every later statement is about those maps.

The most important systems are motions in a potential,

$$\ddot{x} = -\nabla V(x), \qquad E(x,v) = \tfrac{1}{2}\,|v|^2 + V(x), \tag{1.9}$$

with unit masses. The energy is constant on every motion:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}E = \dot{x}\cdot\big(\ddot{x} + \nabla V(x)\big) = 0. \tag{1.10}$$

Proposition 1.9 (Mechanical completeness)**.** *Let $V \in C^2(\mathbb{R}^n)$ be defined on all of $\mathbb{R}^n$ and bounded below. Then every solution of $\ddot{x} = -\nabla V(x)$ exists for all $t \in \mathbb{R}$; the flow on $\mathcal{M} = \mathbb{R}^{2n}$ is complete.*

Proof. The field $F(x,v) = (v, -\nabla V(x))$ is C^1, so T1 applies. Let $V \geq V_{\min}$ and fix a solution with energy E. By (1.10),

$$\tfrac{1}{2}\,|\dot{x}(t)|^2 = E - V(x(t)) \leq E - V_{\min}, \qquad \text{so} \qquad |\dot{x}(t)| \leq C := \sqrt{2(E - V_{\min})}$$

for as long as the solution exists. Integrating, $|x(t)| \leq |x(0)| + C\,|t|$. Suppose $T_+ < \infty$. Then for $0 \leq t < T_+$ the state $(x(t), \dot{x}(t))$ lies in the compact set

$$K = \big\{(x,v) : |x| \leq |x(0)| + CT_+,\ |v| \leq C\big\} \subset \mathbb{R}^{2n},$$

which contradicts the blow-up alternative. So $T_+ = \infty$, and the same argument with t decreasing gives $T_- = -\infty$. ∎

Energy bounds the speed, and a bounded speed cannot reach infinity in finite time. Neither hypothesis can be dropped.

Example 1.10 (The domain must be all of $\mathbb{R}^n$)**.** Take $V = 0$ on $U = (0,\infty)$: a free particle that is only allowed to be at positive x. The solution $x(t) = x_0 - wt$ $(x_0, w > 0)$ reaches the boundary at $t = x_0/w$: $T_+ = x_0/w$, although the energy is bounded. The physical version is the Kepler problem with the collision $\mathbf{r} = 0$ removed: a particle falling straight at the centre reaches it in finite time.

Example 1.11 (The potential must be bounded below)**.** Take $V = -x^4/4$ on $\mathbb{R}$, so $\ddot{x} = x^3$. From the state $(x_0, x_0^2/\sqrt{2})$, $x_0 > 0$, of energy 0, the particle moves outwards with $\dot{x} = x^2/\sqrt{2}$, and

$$x(t) = \frac{x_0}{1 - x_0 t/\sqrt{2}},$$

which reaches infinity at $t = \sqrt{2}/x_0$. The speed $x^2/\sqrt{2}$ is unbounded: the first line of the proof fails.

1.4 One degree of freedom: the energy method

Now let $n = 1$, $V \in C^2(\mathbb{R})$. Orbits lie in level sets of $E = \frac{1}{2}v^2 + V(x)$, so the *phase portrait* can be drawn from the graph of V without solving anything.

Reading the portrait from the potential

On the level E, $v = \pm\sqrt{2(E - V(x))}$, so:

- the motion is confined to the *allowed region* $\{V \le E\}$;
- over an interval of $\{V < E\}$ the level is two graphs, and the orbit runs rightwards on top, leftwards underneath: clockwise;
- at a *turning point* ($V = E$, $V' \neq 0$) the graphs join and the particle reverses;
- the equilibria are the points $(x_*, 0)$ with $V'(x_*) = 0$.

So (Figure 1.2): over a bounded interval of $\{V < E\}$ ending in two turning points, the two graphs join into one periodic orbit (Proposition 1.12 below). A non-degenerate minimum ($V'' > 0$) is surrounded by closed curves: a *centre*. A non-degenerate maximum ($V'' < 0$) is a crossing of its level set: a *saddle*, whose level is a *separatrix* between motions of different kinds. Degenerate critical points need separate analysis.

Proposition 1.12 (One periodic orbit between two turning points)**.** *Let V be C^2 on an open interval J, and let $x_- < x_+$ in J be turning points of the level E: $V(x_\pm) = E$, $V'(x_\pm) \neq 0$, and $V < E$ on (x_-, x_+). Then the two graphs $v = \pm\sqrt{2(E - V)}$ over $[x_-, x_+]$ form a single periodic orbit, with period*

$$T(E) = 2\int_{x_-}^{x_+} \frac{\mathrm{d}x}{\sqrt{2\big(E - V(x)\big)}}. \tag{1.11}$$

Proof. Start at $(x_-, 0)$. There $\ddot{x} = -V'(x_-) > 0$, since $V < E$ just to the right of x_- forces $V'(x_-) \le 0$ and $V'(x_-) \neq 0$. While $x_- < x < x_+$ we have $v = \sqrt{2(E - V(x))} > 0$, so x increases and reaches x at time $\int_{x_-}^{x} \mathrm{d}\xi/\sqrt{2(E - V(\xi))}$. This converges at x_+, because $V'(x_+) > 0$ (by the same argument) gives $E - V(\xi) \ge c(x_+ - \xi)$ nearby, for some $c > 0$, and similarly at x_-. So the particle reaches $(x_+, 0)$ at time $\frac{1}{2}T(E)$. The equation is unchanged by reversing time, so $x(T(E) - t)$ is also a solution, and it agrees with $x(t)$ at $t = \frac{1}{2}T(E)$ with the same velocity 0; by uniqueness they coincide, and the particle is back at $(x_-, 0)$ at time $T(E)$, having run along both branches $v = \pm\sqrt{2(E - V)}$ over $[x_-, x_+]$. ∎

Near a minimum: periodic orbits and stability

Definition 1.13 (Lyapunov stability)**.** An equilibrium z_* of $\dot{z} = F(z)$ is *(Lyapunov) stable* if for every neighbourhood N of z_* there is a neighbourhood N' of z_* such that every solution starting in N' stays in N for all $t \ge 0$. It is *asymptotically stable* if, in addition, all solutions starting in some neighbourhood tend to z_* as $t \to \infty$.

Proposition 1.14 (One-degree-of-freedom Dirichlet)**.** *Let $V \in C^2(\mathbb{R})$ and let x_0 be a critical point with $V''(x_0) > 0$. Then*

1. *the equilibrium $(x_0, 0)$ is Lyapunov stable;*
2. *there are $a < x_0 < b$ and $\varepsilon > 0$ such that for $V(x_0) < E < V(x_0) + \varepsilon$ the part of the level*

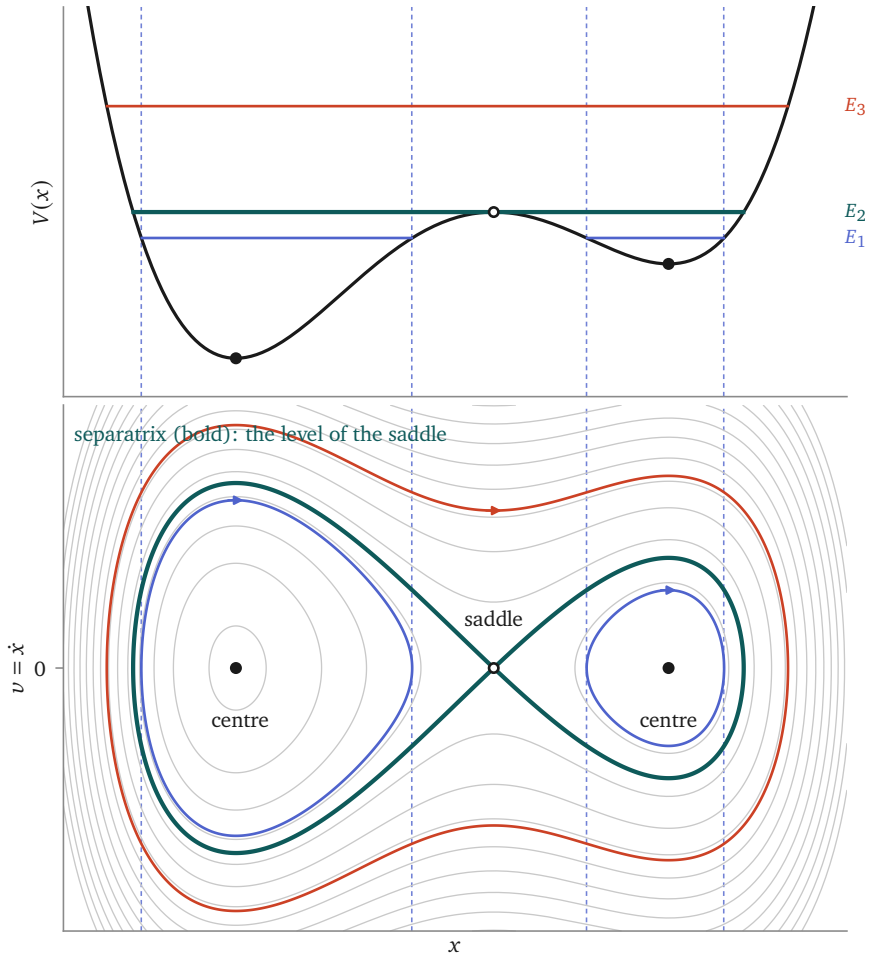


Figure 1.2: Potential over portrait, $V = \frac{1}{4}x^4 - \frac{1}{2}x^2 + \frac{1}{8}x$. The dashed verticals carry the turning points of E_1 down. E_1 carries *two* orbits: the energy does not determine the motion, the state does. E_2 (bold) is three orbits: the saddle and two loops that leave and return to it. E_3 is one orbit round both wells.

> *E over (a,b) is a single periodic orbit, with the period (1.11) between the turning points $x_-(E) < x_0 < x_+(E)$ nearest to x_0.*

Proof. The hypothesis, as used. Since V' is increasing through 0 at x_0, there are $a < x_0 < b$ with $V' < 0$ on $[a, x_0)$ and $V' > 0$ on $(x_0, b]$. Put $\varepsilon = \min(V(a), V(b)) - V(x_0) > 0$.

Trapping sets. For $V(x_0) < E < V(x_0) + \varepsilon$ there are unique $x_-(E) \in (a, x_0)$, $x_+(E) \in (x_0, b)$ with $V = E$, and $V < E$ between them. Let

$$D_E = \left\{(x, v) : x_-(E) \le x \le x_+(E),\ \tfrac{1}{2}v^2 + V(x) \le E\right\}.$$

It is compact and shrinks to $(x_0, 0)$ as $E \downarrow V(x_0)$. A solution starting in D_E never leaves it: its energy stays at most E, so $x(t)$ cannot enter $[a, x_-(E))$ or $(x_+(E), b]$, where $V > E$, and being continuous it cannot jump over them. Given a neighbourhood N of $(x_0, 0)$, choose E with $D_E \subset N$ and let N' be the interior of D_E. This proves (1).

The orbit. $x_\pm(E)$ are turning points of the level E, since $V'(x_-) < 0 < V'(x_+)$, so by Proposition 1.12 the two graphs over $[x_-, x_+]$ form one periodic orbit with period (1.11). They make up the whole level E over (a, b), since $V \le E$ there only on $[x_-, x_+]$. ∎

1.5 The pendulum and the cylinder

A rigid pendulum at angle x from the downward vertical obeys, in units in which the length, mass and gravitational acceleration are 1,

$$\ddot{x} = -\sin x, \qquad V(x) = 1 - \cos x, \qquad E = \tfrac{1}{2}v^2 + 1 - \cos x. \tag{1.12}$$

The flow is complete by Proposition 1.9. V has minima at $x = 2\pi k$ ($E = 0$) and maxima at $x = \pi + 2\pi k$ ($E = 2$), and the portrait follows (Figure 1.3):

- $0 < E < 2$: *libration*, a periodic swing about the bottom, round a centre, as in Proposition 1.14;
- $E = 2$: the *separatrix* through the upright position, which is a saddle;
- $E > 2$: *rotation*, with v never zero, so the pendulum goes over the top again and again in one direction.

The separatrix takes forever

On $E = 2$ the solution through $(0, 2)$ is

$$x(t) = 4\arctan e^t - \pi, \qquad v(t) = 2\operatorname{sech} t. \tag{1.13}$$

As $t \to \pm\infty$, $x \to \pm\pi$ and $v \to 0$: the pendulum takes infinite time to reach the upright position. It must: the saddle is an equilibrium, hence an orbit of its own, and by Corollary 1.6 no other motion can reach it. The separatrix is not a periodic orbit.

The period, and its logarithmic divergence

A libration of energy $0 < E < 2$ runs between the turning points $\pm\arccos(1 - E)$, where $V' = \sin x \neq 0$, so Proposition 1.12 applies. Substituting $\sin\frac{x}{2} = k\sin u$ in (1.11) turns $2(E - 1 + \cos x)$ into $4k^2\cos^2 u$ and $\mathrm{d}x$ into $2k\cos u\,\mathrm{d}u/\sqrt{1 - k^2\sin^2 u}$, and gives

$$T(E) = 4K(k), \qquad K(k) = \int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - k^2\sin^2 u}}, \qquad k^2 = \frac{E}{2}, \tag{1.14}$$

a complete elliptic integral. As $E \to 0$, $T \to 2\pi$. As $E \uparrow 2$ the integrand develops the singularity $1/\cos u$ at $u = \pi/2$, and the expansion $K(k) = \ln(4/k') + o(1)$, $k' = \sqrt{1-k^2}$ (*NIST Digital Library of Mathematical Functions*, 19.12.1), gives

$$T(E) = 2\ln\frac{32}{2-E} + o(1) \qquad (E \uparrow 2). \tag{1.15}$$

The angle is periodic: the cylinder

The angles x and $x + 2\pi$ are the *same* position of the pendulum. The field $(v, -\sin x)$ is 2π-periodic in x, so its flow commutes with $x \mapsto x + 2\pi$, and we may identify (x, v) with $(x + 2\pi, v)$. What is left is the strip $-\pi \le x \le \pi$ with its edges glued: a *cylinder*.

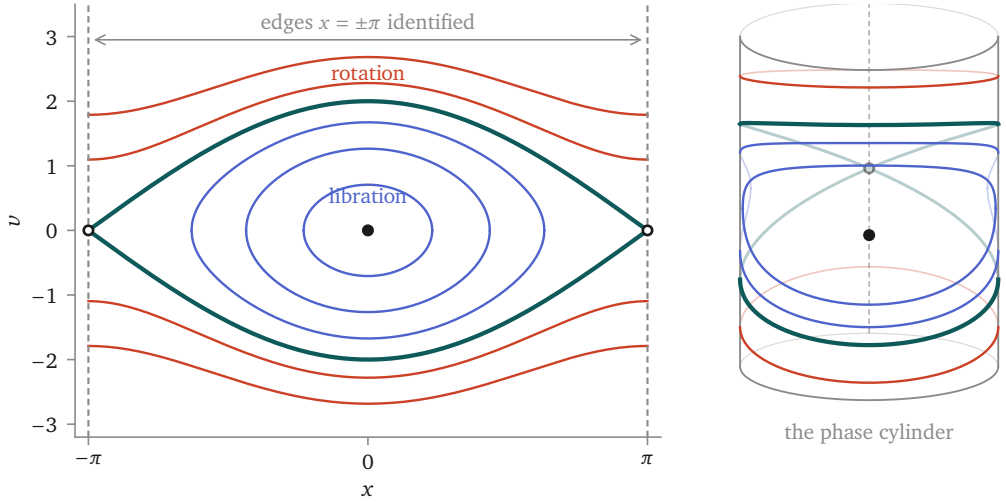


Figure 1.3: The pendulum's portrait on the strip (left) and on the cylinder (right; back half faint). Rotations, which leave the strip through one edge and return through the other, close up into loops round the cylinder.

On the cylinder the rotations become what they physically are: *periodic* motions. In the plane they were injective curves, and "every motion with $E \neq 2$ is periodic" was false; on the cylinder it is true. The complete flow of the plane commutes with the shift, so it defines a complete flow on the cylinder.

This is the first sign that phase spaces are not always open subsets of $\mathbb{R}^{2n}$. We stay in coordinates, remembering that x is defined modulo 2π; Lecture 2 meets curved configuration spaces.

1.6 Why phase space, and a contrast

Why phase space

In configuration space, or in configuration space and time, curves cross: the motions of Figure 1.1 meet again at $t = \pi$, and a pendulum passes the bottom going both ways. In phase space they cannot: by Corollary 1.6 through every state there is *exactly one* orbit. That is determinacy made geometric, and it is why a phase portrait is a family of non-intersecting curves while trajectories in space are a tangle.

For $\ddot{x} = -\nabla V$ with V as in Proposition 1.9, the course's three questions now have provisional answers:

state	a point $z = (x, v)$ of $\mathcal{M} = \mathbb{R}^{2n}$
observable	a function $f\colon \mathcal{M} \to \mathbb{R}$
evolution	the flow φ_t: states move by $z \mapsto \varphi_t(z)$, observables by $f \mapsto f \circ \varphi_t$

Lecture 7 and Lecture 12 return to this table with the entries rebuilt.

A contrast: friction

Add friction to the pendulum:

$$\ddot{x} = -\sin x - \gamma \dot{x}, \qquad \gamma > 0. \tag{1.16}$$

T1 and T2 apply unchanged, but now

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \dot{x}\,\ddot{x} + \sin x\,\dot{x} = -\gamma\,\dot{x}^2 \le 0,$$

and orbits cross the energy levels downhill (Figure 1.4). The linearisation at $(0,0)$, $\left(\begin{smallmatrix} 0 & 1 \\ -1 & -\gamma \end{smallmatrix}\right)$, has trace $-\gamma < 0$ and determinant 1, so both eigenvalues have negative real part. By the linearisation theorem (Teschl, Thm 6.10, cited and not needed again) the bottom equilibria are asymptotically stable: *attractors*.

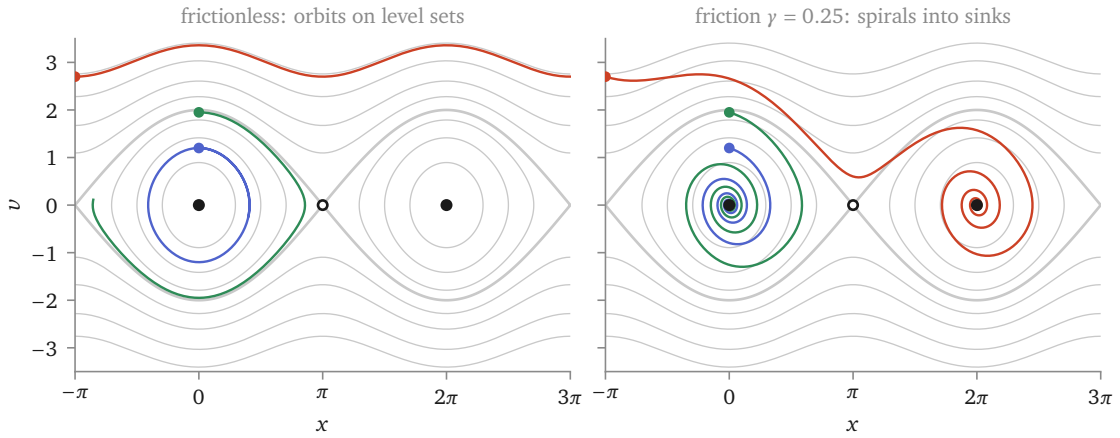


Figure 1.4: Three initial states without friction (left) and with $\gamma = 0.25$ (right); faint: the levels of E. With friction each plotted orbit spirals into a sink.

Without friction the minimum is stable (Proposition 1.14) but never asymptotically stable, and that is not an accident of one degree of freedom.

Question. *What property of the flows of mechanics — as opposed to the damped pendulum — forbids attractors?*

Energy conservation already rules out asymptotically stable equilibria: a nearby state of different energy can never reach the equilibrium's energy. It says less about other attractors, such as limit cycles. A better hint is area. The frictionless flow shears a blob of initial states but keeps its area; the damped flow shrinks it, at the rate given by the divergence of the field, 0 against $-\gamma$. Lecture 7 answers the question.

Summary.

- *Handed on to every lecture:* the state $z = (x, \dot{x})$ and the flow φ_t of $\dot{z} = F(z)$, with its group law, and orbits that never meet (Definition 1.1, Theorem 1.5, and Corollary 1.6); completeness, the standing hypothesis, which holds for $V \in C^2(\mathbb{R}^n)$ bounded below (Proposition 1.9).
- *In one degree of freedom:* one periodic orbit between two turning points, and a stable centre at a non-degenerate minimum (Propositions 1.12 and 1.14).
- *The running example:* the pendulum, its separatrix, its period $T = 4K(k)$ and its cylinder.
- *Next:* Lecture 2 puts constraints on the configuration. The damped pendulum's question — what forbids attractors? — is answered in Lecture 7.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §1 (determinacy) and §4 (one degree of freedom, pp. 15–22). Teschl, *Ordinary Differential Equations and Dynamical Systems*, Chs. 2 and 6, for T1, T2 and T5, and §6.7 for a careful treatment of the pendulum. Hirsch, Smale and Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, Chs. 7 and 9, for the example of § 1.2 and the damped pendulum. *NIST Digital Library of Mathematical Functions*, §19.12, for the elliptic integral near $k = 1$.